



Lecture 25:

Introduction to TDR & VNA

Contents



This Topic: Introduction TDR & VNA

- Introduction to TDR
- Introduction to VNA

Introduction



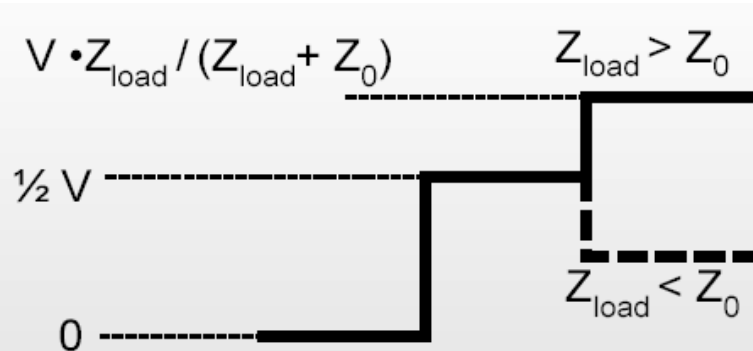
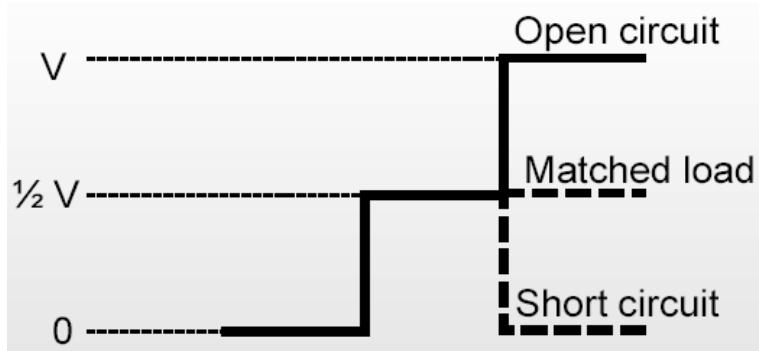
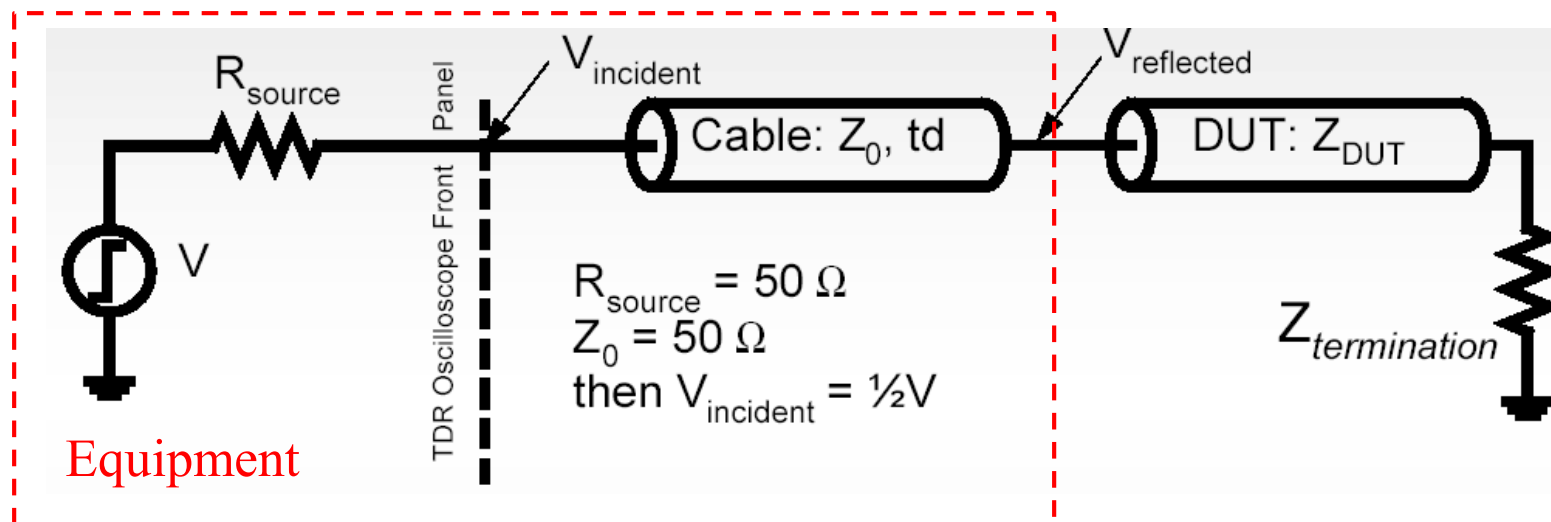
- Time Domain Reflectometry (TDR) is a very useful lab method to characterize the channel or load
- TDR measures the reflections from a signal traveling through a circuit
 - circuit board trace
 - cable,
 - connector and so on.
- TDR sends a pulse through the medium and compares the reflections of the channel/load with a standard impedance.

Introduction

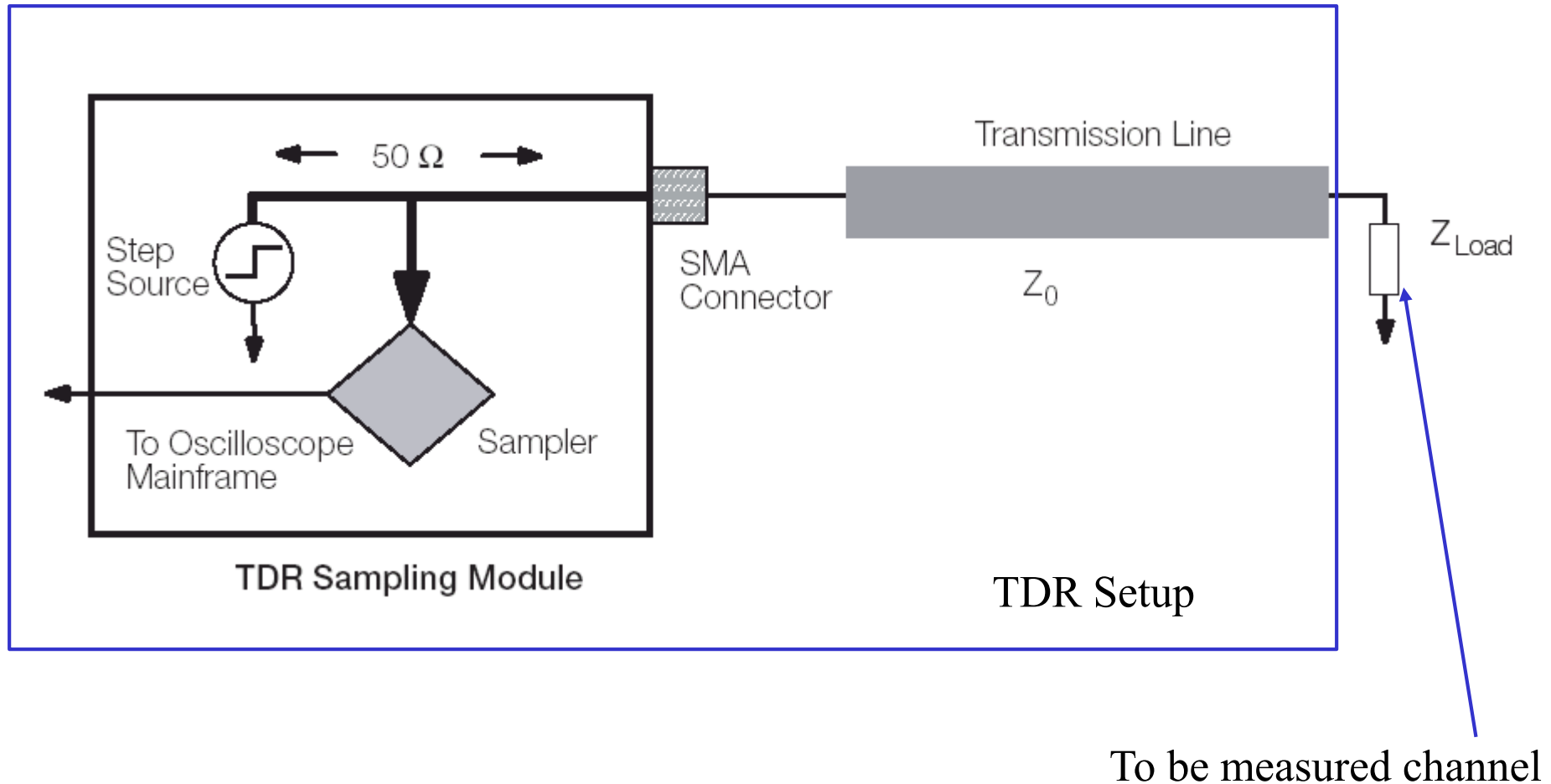


- TDR display is the voltage waveform that returns when a fast step signal is propagated down a transmission line.
- The resulting waveform is the combination of the incident step and reflections generated when the step encounters impedance variations.

TDR Measurement Basics



TDR Measurement Setup



TDR Measurement Basics

- TDR measurements are based on a series of impedance ratios. Most TDR instruments will perform the necessary ratio calculations internally and display a numerical result.

$$\rho = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{(Z_L - Z_0)}{(Z_L + Z_0)}$$

$$Z_L = Z_0 * \frac{(1 + \rho)}{(1 - \rho)}$$

Units of
Magnitude

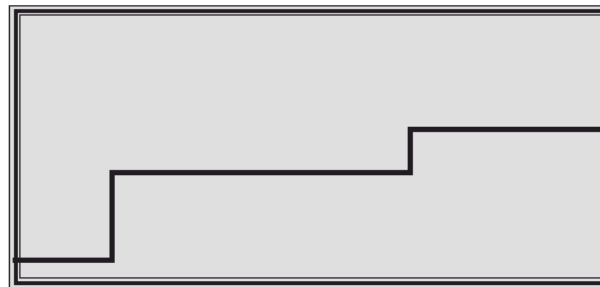
Volts/Div

-or-

Rho/Div

-or-

Ohms/Div

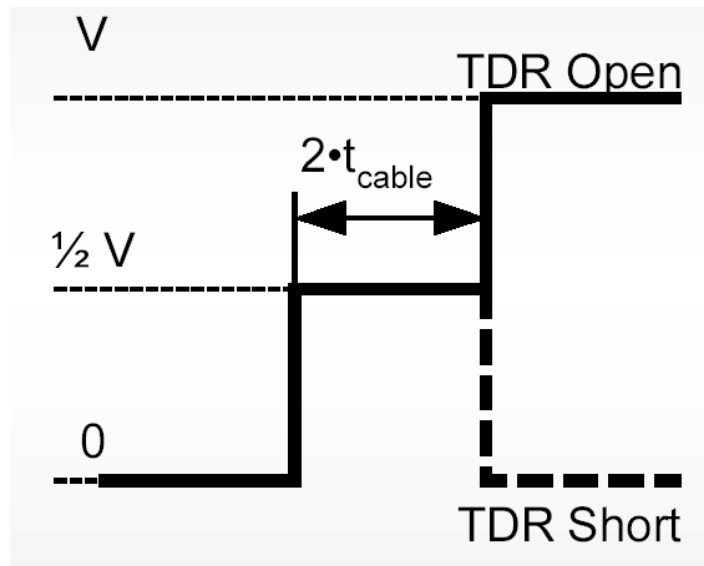


Units of Time - Sec/Div

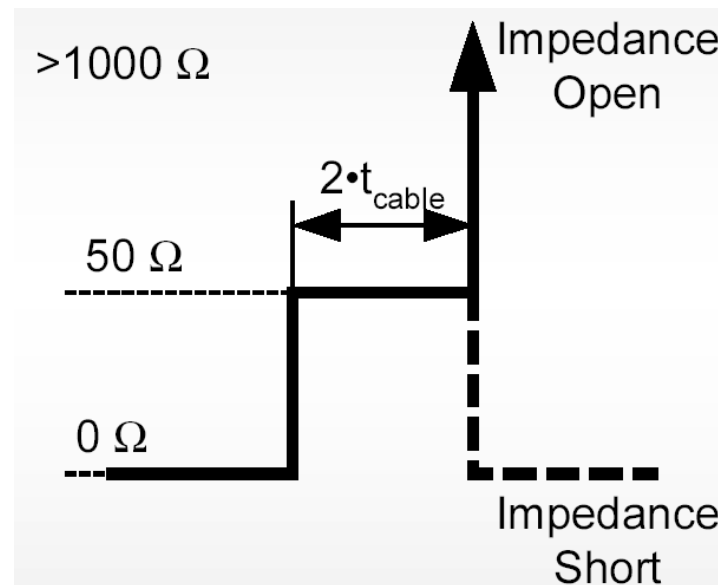
Simplified diagram of a TDR waveform

EEE598 Spring 2025

Electrical Signatures



Reflection Profile

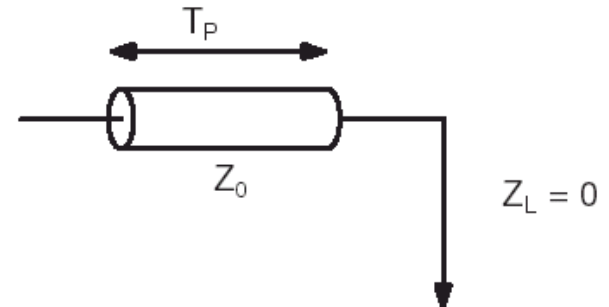
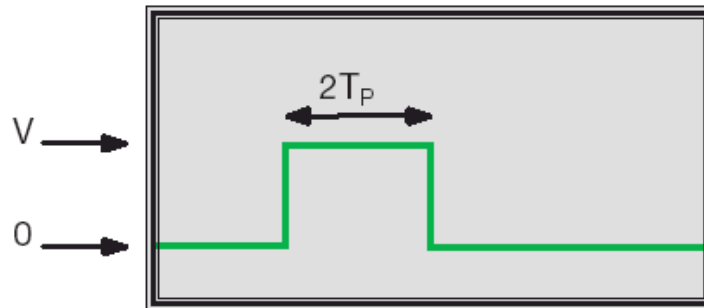


Impedance Profile

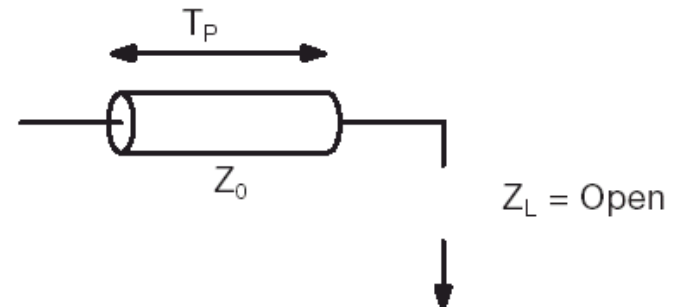
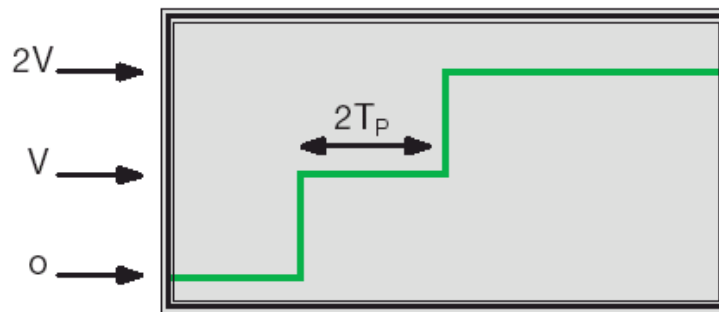
Electrical Signatures (Short and Open)



– Short Circuit Termination



– Open Circuit Termination

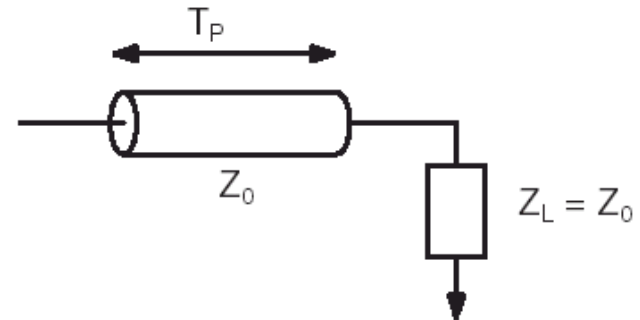
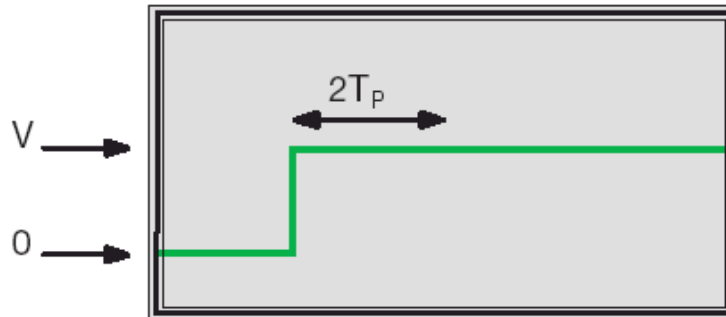


Short and open circuit terminations

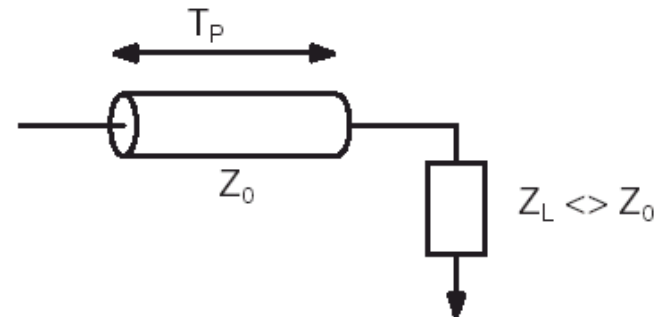
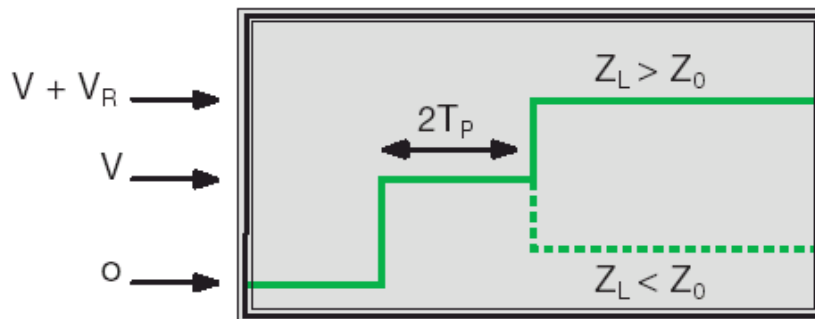
Electrical Signatures (R Components)



– Matched Load Termination



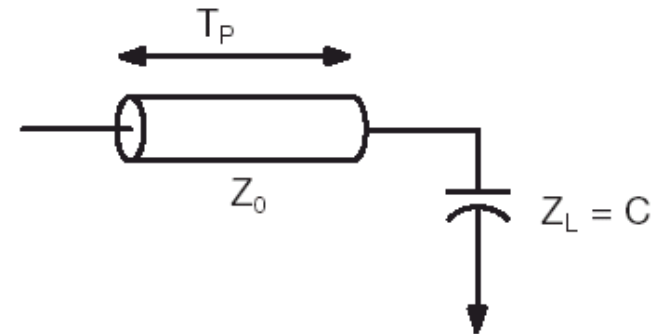
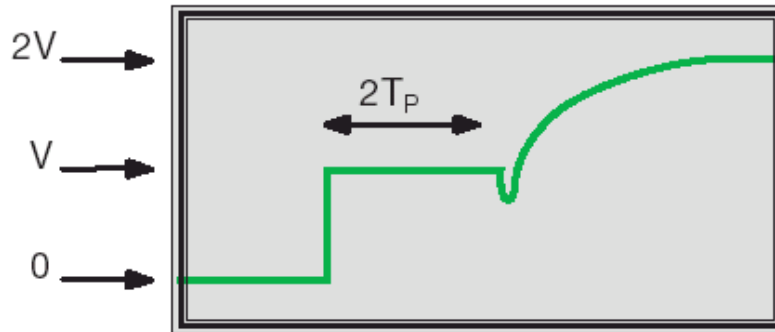
– Mismatched Load Termination



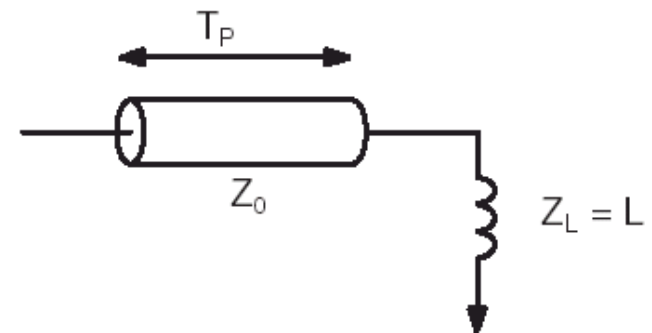
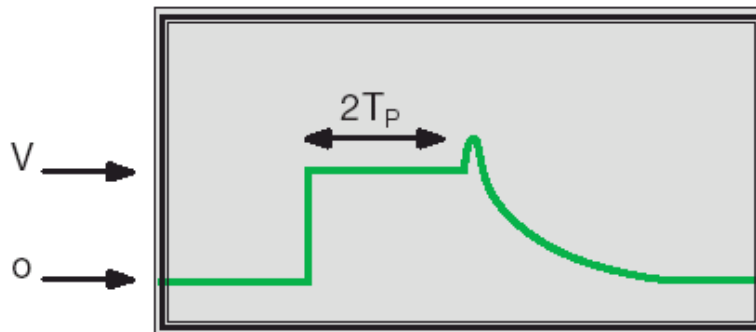
Matched and mismatched load terminations

Electrical Signatures (C and L Components)

– Capacitor Load Termination



– Inductor Load Termination

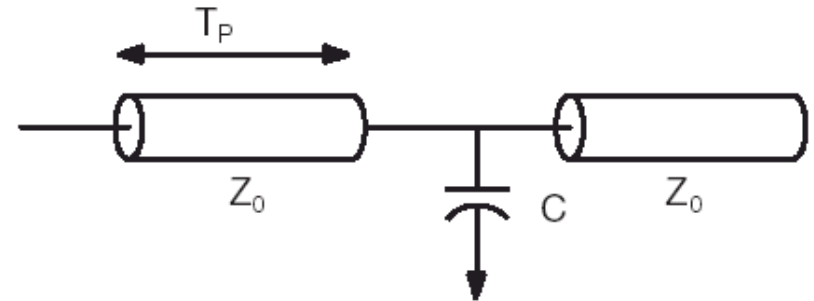
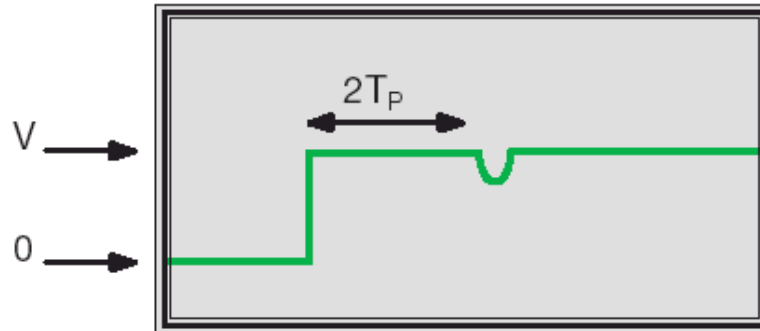


Capacitive and inductive load terminations

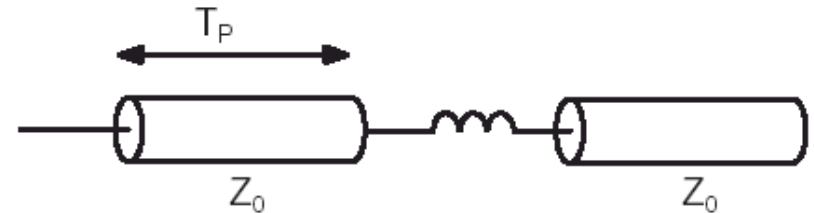
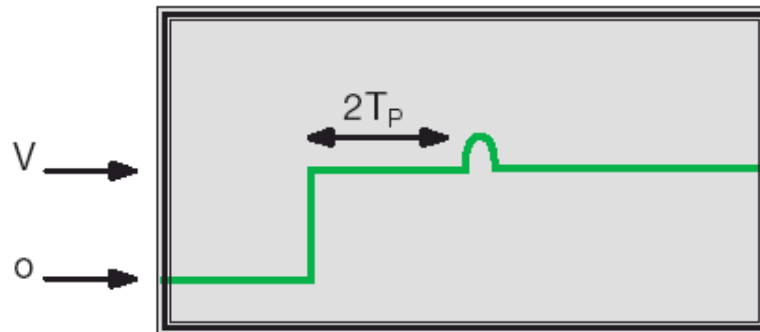
Electrical Signatures (C and L Discontinuity)



– Shunt Capacitance Discontinuity



– Series Inductance Discontinuity

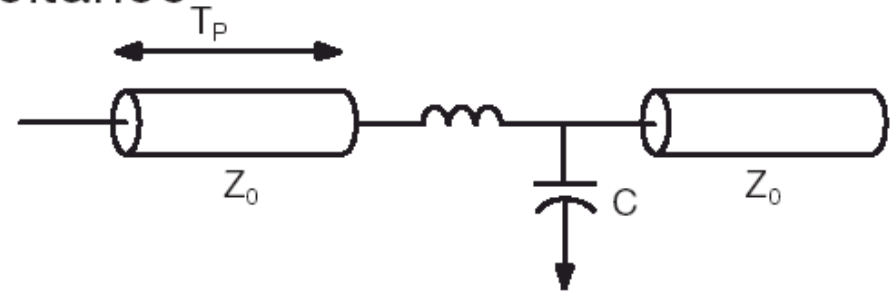
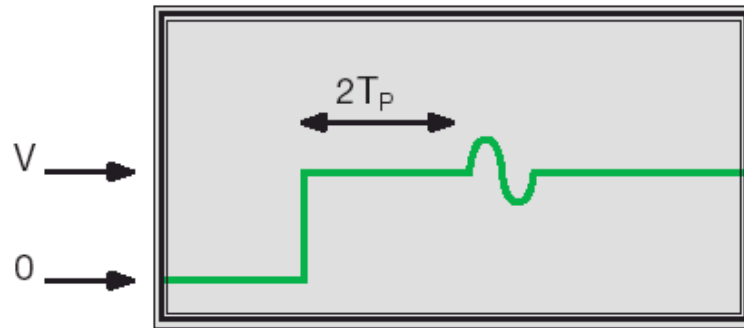


Capacitive and inductive discontinuities

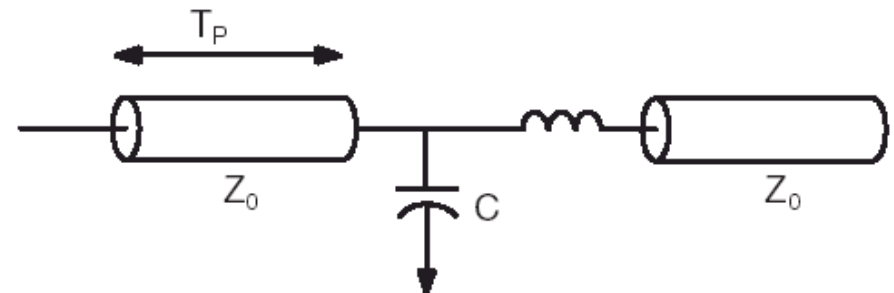
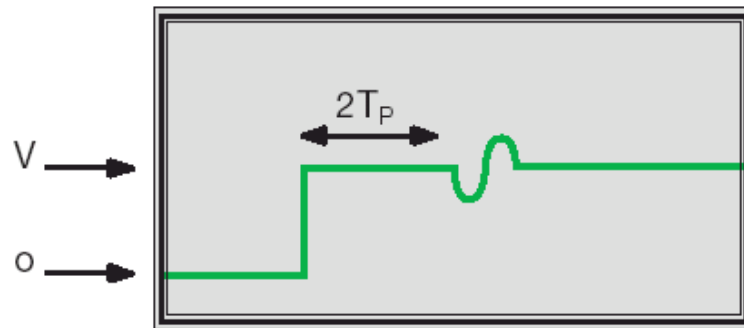
Electrical Signatures (C and L Discontinuity)



– Series Inductance – Shunt Capacitance

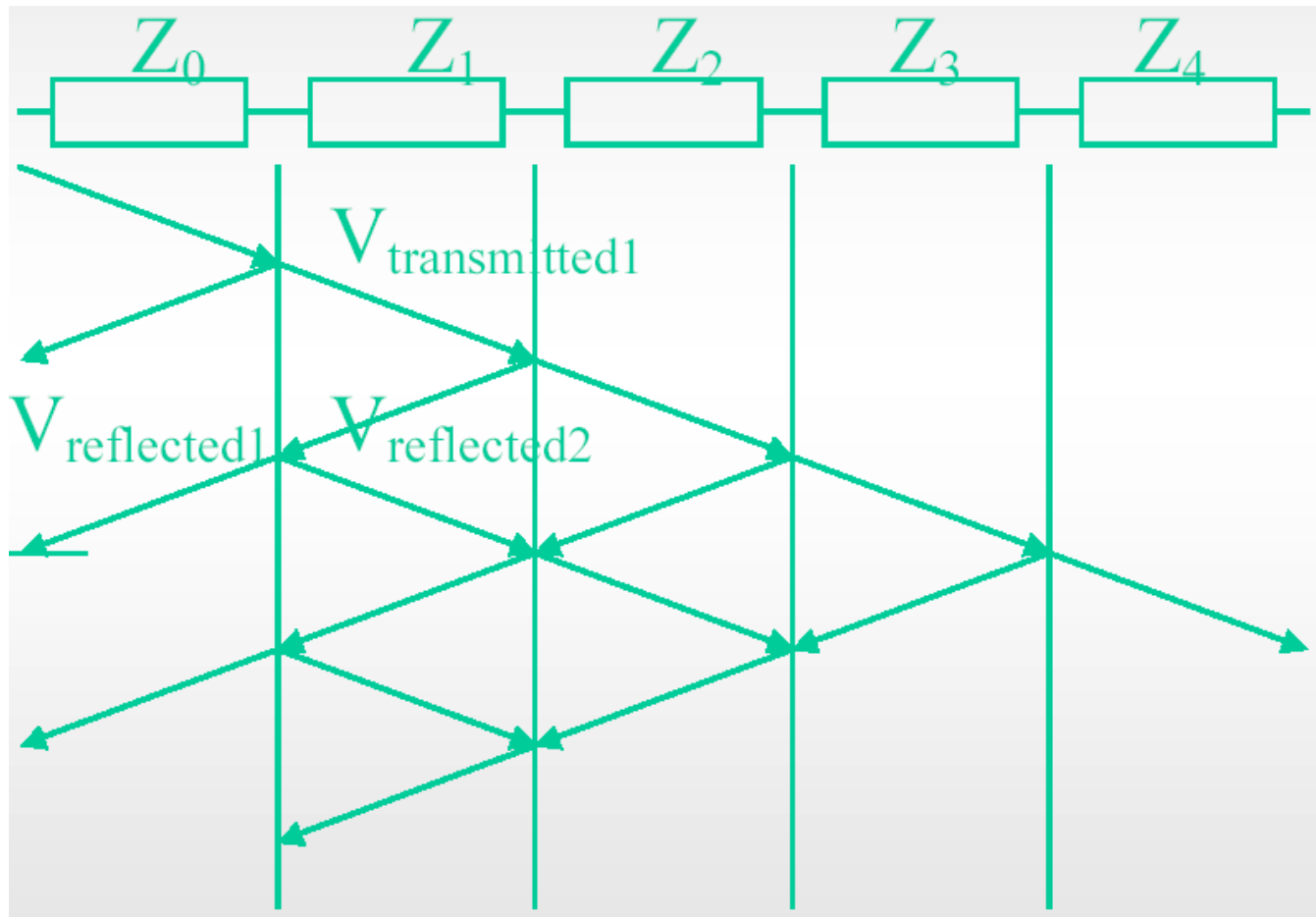


– Shunt Capacitance – Series Inductance

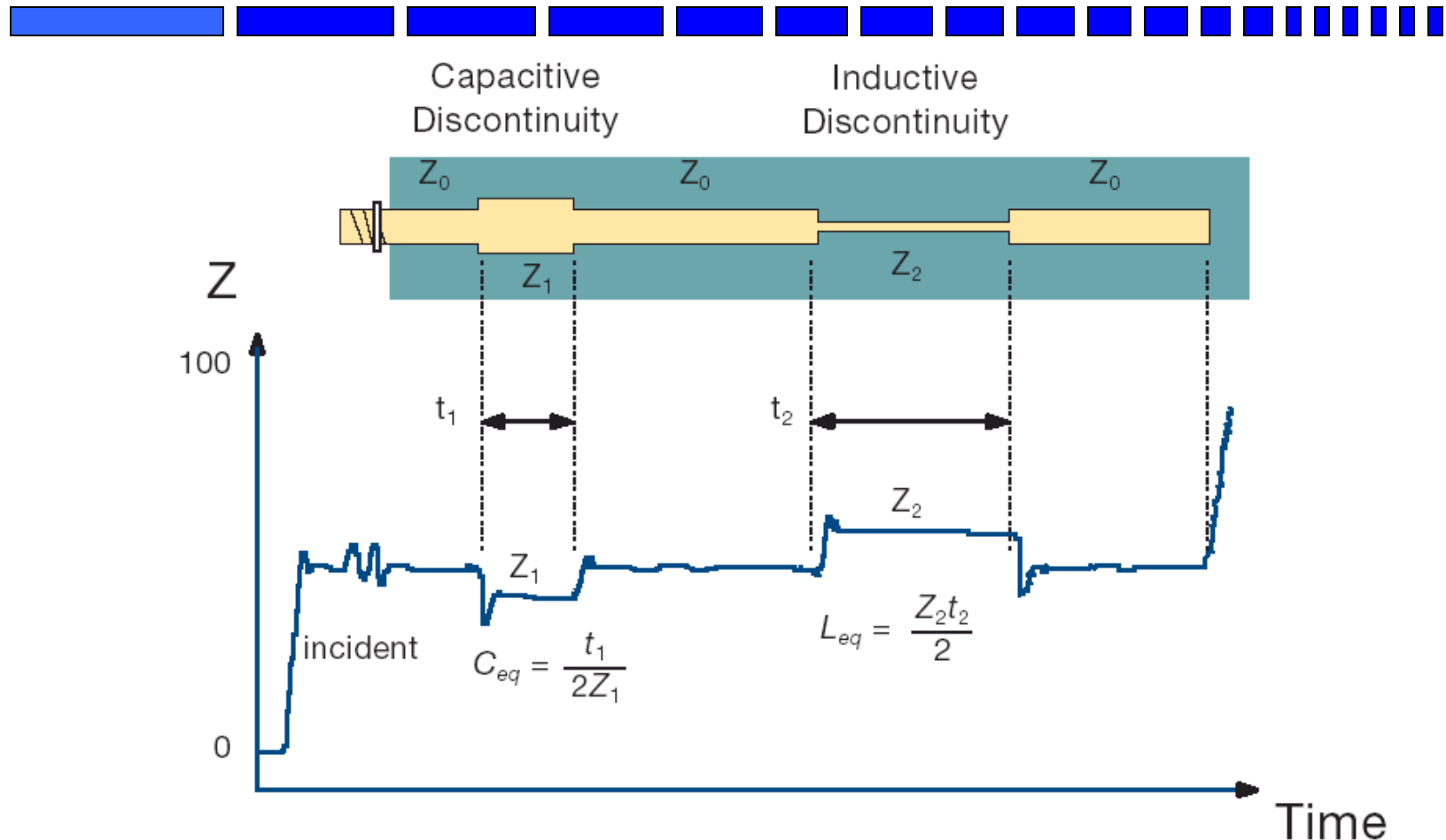


Mixed capacitive and inductive loading

Multiple TDR Reflection Effects



Electrical Signatures (Trace Discontinuity)



The TDR waveform reveals trace discontinuities

Introduction to Vector Network Analyzer (VNA)

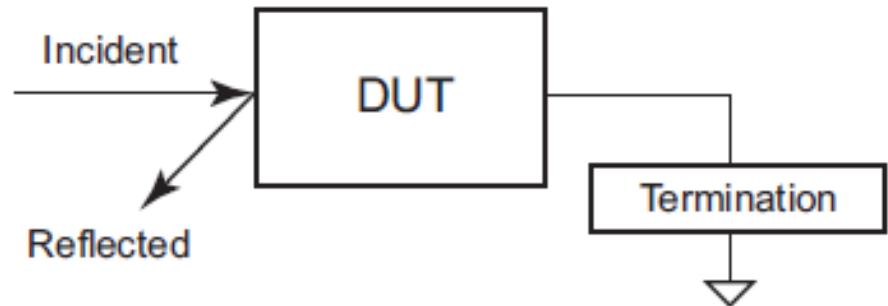


Transmission Measurements



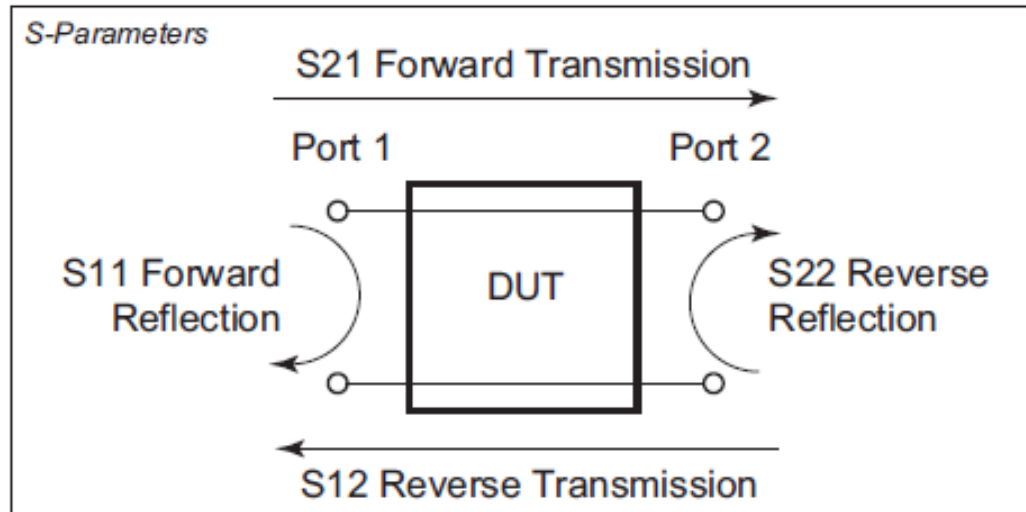
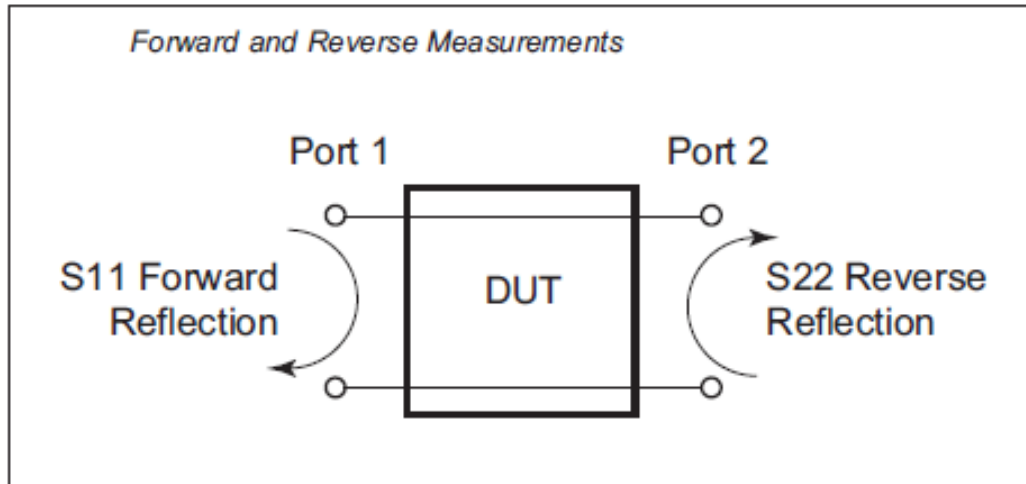
- Gain (dB)
- Insertion Loss (dB)
- Insertion Phase (degrees)
- Transmission Coefficients (S_{12} , S_{21})
- Separations of Transmission Components (Real and Imaginary)
- Electrical Length (m)
- Electrical Delay (s)
- Deviation from Linear Phase (degrees)
- Group Delay (s)

Reflection Measurements



- Return Loss (dB)
- Reflection Coefficients (S_{11} , S_{22})
- Reflection Coefficients vs Time (Fourier Transform)
- Impedance ($R \pm jX$)
- SWR

Introduction to Vector Network Analyzer (VNA)



Reading Assignment



- Please read Chapter 20 of the textbook

VLSI High-Speed I/O Circuits

Theoretical Basis, Architecture, Modeling and Circuit Implementation



VLSI High-Speed I/O Circuits

By: Hongjiang Song

ISBN10: 1-4415-5987-6 (Trade Paperback 6x9)

ISBN13: 978-1-4415-5987-6 (Trade Paperback 6x9)

ISBN10: 1-4415-5988-4 (Trade Hardback 6x9)

ISBN13: 978-1-4415-5988-3 (Trade Hardback 6x9)

Pages : 489

Book Format : Trade Book 6x9

Subject : TECHNOLOGY / Engineering / Electrical
TECHNOLOGY / Engineering / General

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